YILDIZ TECHNICAL UNIVERSITY
CIVIL ENGINEERING DEPARTMENT
MECHANICS DIVISION

STRENGTH OF MATERIALS II

COMBİNED LOADING

FIRST CHAPTER

NORMAL FORCE AND BENDING

1.5 Weeks, First in Line

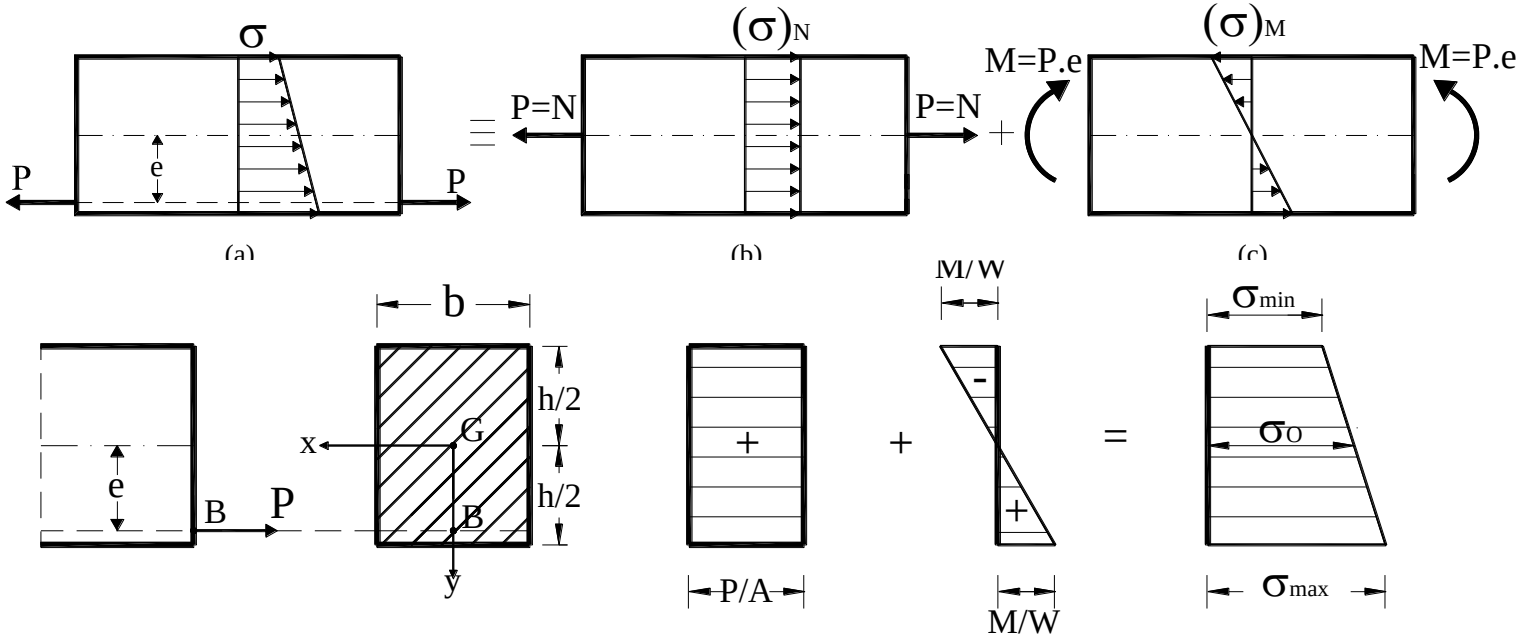
PROF. DR. TURGUT KOCATÜRK

1- Eccentric Normal Force

Axial normal force (N) and bending moment (M) acting together in the cross-section of the beam occurs in the case of eccentric normal force.

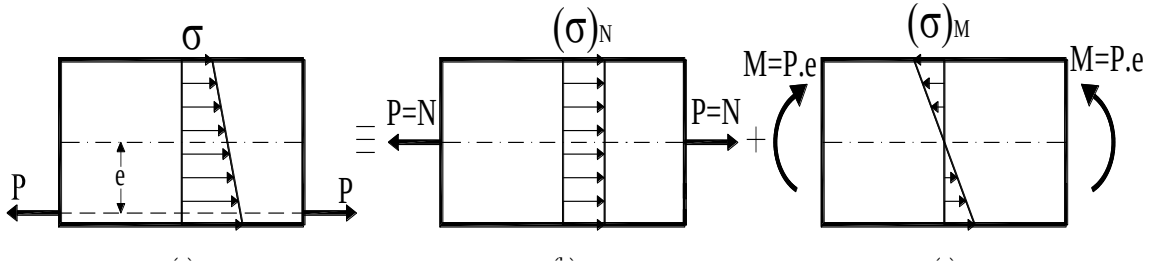
If the normal force acting on the beam is not on the axis of the beam, it is called the eccentric normal force case. Such a force when brought to the axis of the beam always produces an *axial normal force and a bending moment.*

As shown in the figure below, "e" is the distance between applied force point and the axis of the beam. When we move the force P to the axis of the beam, this process produces the bending moment P.e and the normal force P.



If the eccentric normal force is reduced to two simple states, then the stress can be calculated by using pure bending and normal force states. Although the stress in the case of axial normal force is uniformly distributed through the cross-section, in the case of pure bending moment, stresses have two signs as tensile stresses on one side of the cross-section and compressive stresses on the other side of the cross section.

Bending stresses + axial normal stresses = Single or double sign of stress distribution depending on the eccentricity.

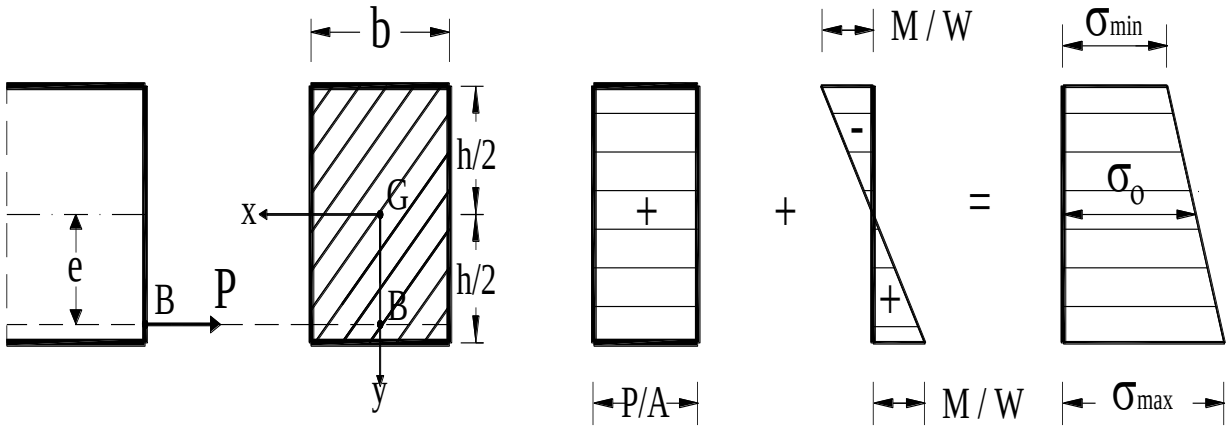


$$\sigma = \frac{N}{A} + \frac{M}{I_x}.y \qquad \sigma = \frac{P}{bh} + \frac{12Pe}{bh^3}.y$$

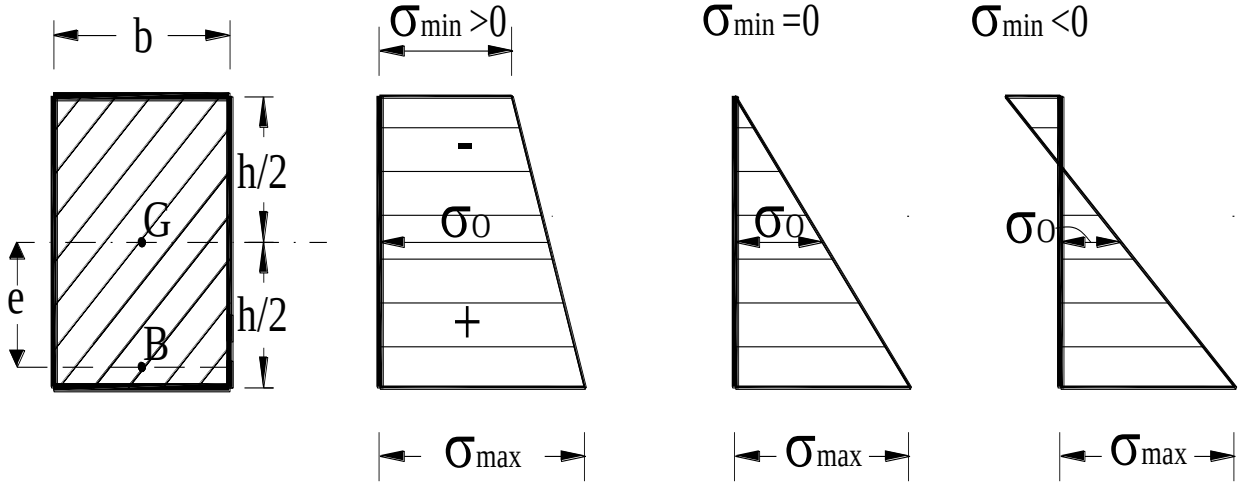
$$y = \pm h/2 \text{ for}$$

$$\sigma_{max} = \frac{P}{bh}\left(1 + \frac{6e}{h}\right)$$

$$\sigma_{min} = \frac{P}{bh}\left(1 - \frac{6e}{h}\right)$$

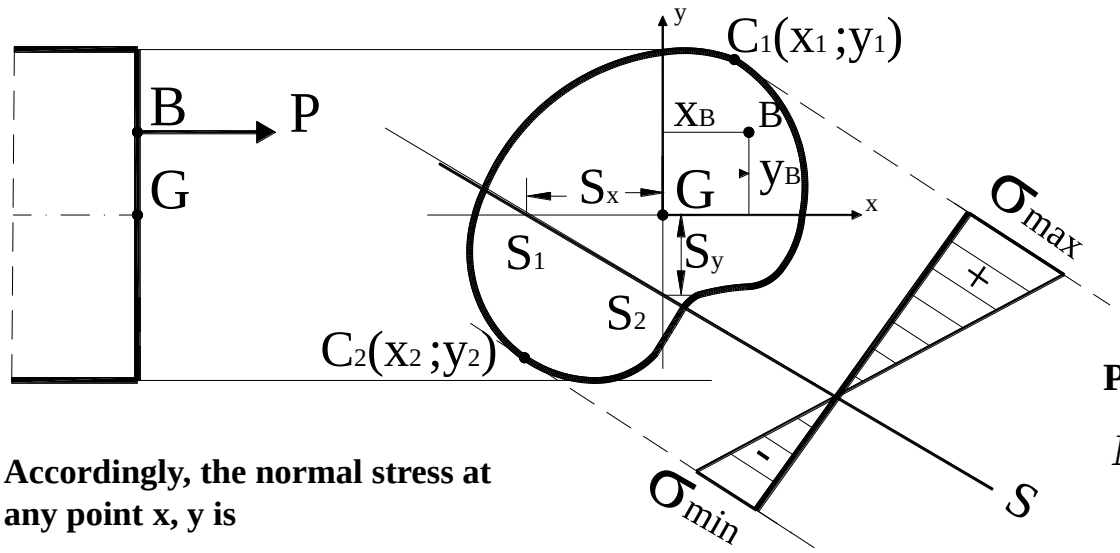


Stresses on neutral axis are always, $\sigma_0 = P/A$ **, not depending on the eccentricity. Stress** σ_{min} **could be tension, zero, or compression**



2- General Calculation of State of Stress

In the previous section, an investigation was made for the special section of rectangular shape, and now the investigation will be made for an arbitrary cross-section (Figure 4).



x, y: principal axes

G: center of gravity

S-S: neutral axis

P eccentric force brought to the point B:

$$N = P;\ M_x = P.y_B;\ M_y = -P.x_B$$

Accordingly, the normal stress at any point x, y is

$$\sigma_z(x,y) = \frac{N}{A} + \frac{M_x}{I_x}y - \frac{M_y}{I_y}x = \frac{P}{A} + \frac{P.y_B.y}{I_x} + \frac{P.x_B.x}{I_y}$$

There is need to determine the position of the neutral axis.

From $\sigma(x,y) = 0$ **, the equation of the neutral axis can be written as follows:**

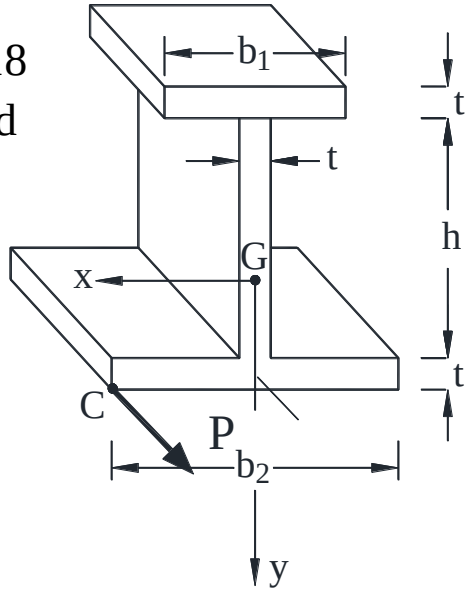
$$\sigma_z(x,y) = \frac{P}{A} + \frac{P.y_B.y}{I_x} + \frac{P.x_B.x}{I_y} = \frac{P}{A}\left(1 + \frac{y_B y}{i_x^2} + \frac{x_B x}{i_y^2}\right) = 0$$

$$i_x^2 = \frac{I_x}{A} \quad \textbf{and} \quad i_y^2 = \frac{I_y}{A} \qquad S_x = -\frac{i_y^2}{x_B} \quad \textbf{and} \quad S_y = -\frac{i_x^2}{y_B}$$

$$\sigma_{max} = \frac{P}{A} + \frac{Py_B}{I_x}.y_1 + \frac{Px_B}{I_y}.x_1$$

$$\sigma_{min} = \frac{P}{A} + \frac{Py_B}{I_x}.y_2 + \frac{Px_B}{I_y}.x_2$$

Problem: P=170 kN eccentric tensile force is acting at point C of the cross-section given below. Cross-sectional dimensions are b_1=12 cm, b_2=18 cm, h=18 cm, and t=2 cm, determine the max. and min. normal stresses at the section and draw the normal stress diagram.



Solution:

$$y_G = \frac{12.2.21 + 18.2.11 + 18.2.1}{12.2 + 2.18.2} = 9.75\ cm$$

$$I_x = \frac{18.2^3}{12} + 18.2.(1-9.75)^2 + \frac{2.18^3}{12} + 2.18.(11-9.75)^2 + \frac{12.2^3}{12} + 12.2.(21-9.75)^2 = 6842\ cm^4$$

$$I_y = \frac{2.12^3}{12} + \frac{2.18^3}{12} + \frac{18.2^3}{12} = 1272\ cm^4, \quad A = 96\ cm^4$$

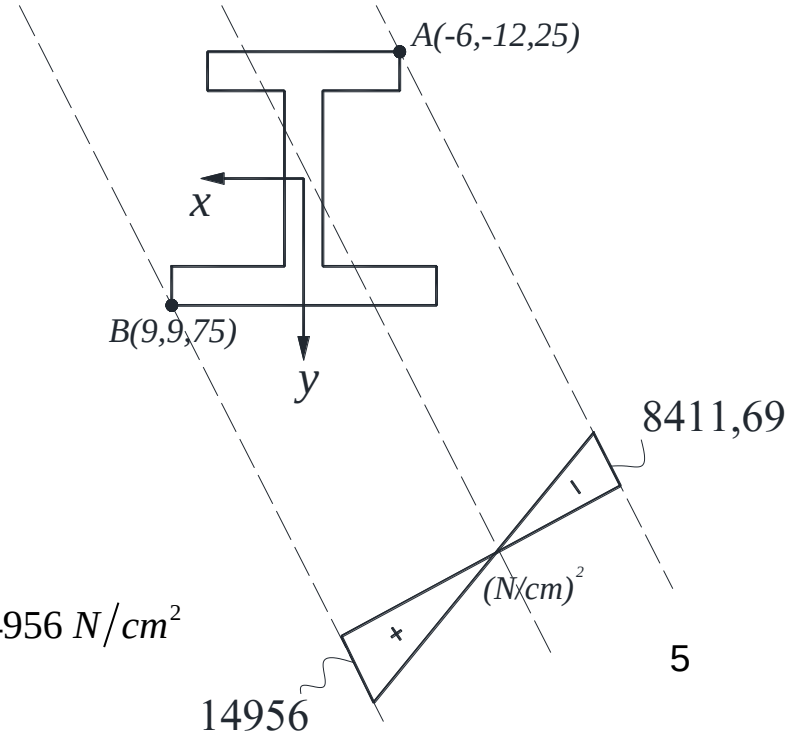
$$M_x = 170000.9,75 = 1657000\ Ncm$$

$$M_y = -170000.9 = -1530000\ Ncm$$

$$\sigma_z = \frac{170000}{96} + \frac{1657000}{6842}y + \frac{1530000}{1272}x \Rightarrow From\ \sigma_z = 0 \Rightarrow$$

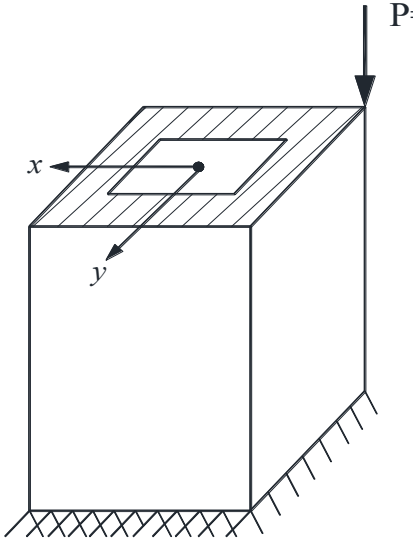
$$y = -4.96x - 7.314\ (Neutral\ axis)$$

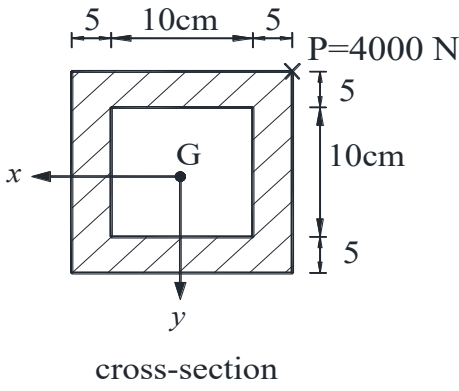
$$\sigma_z^A(-6;12,25) = \sigma_{\min} = -8411.69\ N/cm^2, \quad \sigma_z^B(9;9,75) = \sigma_{\max} = 14956\ N/cm^2$$



Problem: A column with known dimensions is under compression of eccentric load, *P=4000 N* as seen in the figure. Draw the normal stress distribution over cross section of the column.

$$\sigma_z(x,y) = \frac{N}{A} + \frac{M_x}{I_x}y - \frac{M_y}{I_y}x = \frac{P}{A} + \frac{P.y_B.y}{I_x} + \frac{P.x_B.x}{I_y}$$





cross-section

Solution:

$P = -4000N$

$M_x = +4000.10 = +40000Ncm$

$M_y = -4000.10 = -40000Ncm$

$$I_x = I_y = \frac{20.20^3}{12} - \frac{10.10^3}{12} = 12500cm^4$$

$A = 20^2 - 10^2 = 300cm^2$

$$\sigma_z = \frac{-4000}{300} + \frac{40000}{12500}y - \frac{-40000}{12500}x$$

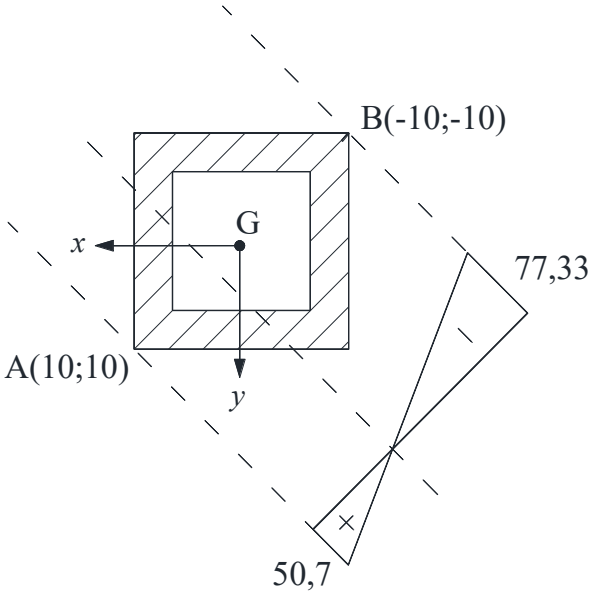
$\sigma_z = 0 \Rightarrow y = -x + 4{,}156 \Rightarrow$

neutral axis:

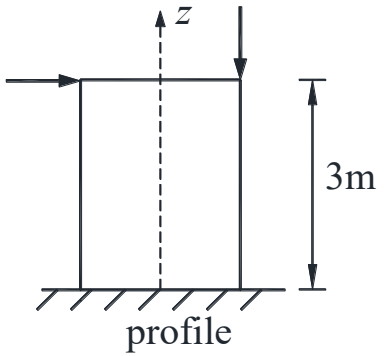
$(0;4{,}156),\ (4{,}156;0)$

$(\sigma_z)_A = 50{,}77N/cm^2$

$(\sigma_z)_B = -77{,}33N/cm^2$



Problem: P and Q forces are acting on the square cross section of the column as seen in the figure. Determine the neutral axis and draw the normal stress distribution of the cross section at the base of the column.



Solution

$$N = -200kN$$

$$M_x = 0$$

$$M_y = -3.50 - 0,5.200 = -250kNm$$

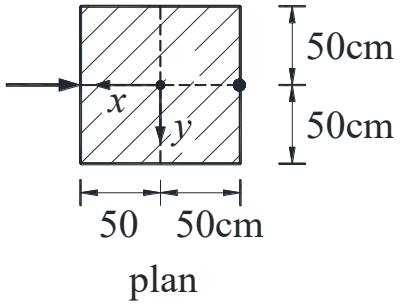
$$A = 1m^2 \qquad I = \frac{1.1^3}{12} = 0,0833m^4$$

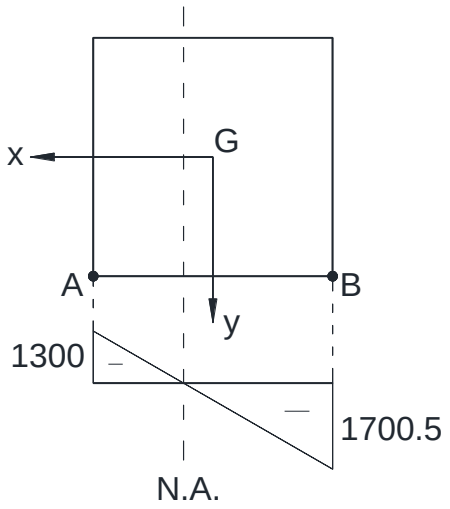
$$\sigma_z = \frac{-200}{1} + \frac{250}{0,0833}x \quad \Rightarrow \quad \sigma_z = -200 + 3001x$$

$$\sigma_z = 0 \quad \Rightarrow \quad x = +0,066$$

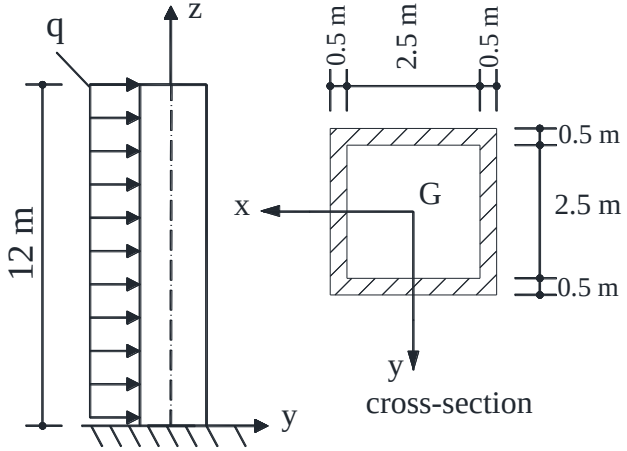
$$(\sigma_z)_A = -200 + 3001.0,5 = 1300\,kN/m^2$$

$$(\sigma_z)_B = -200 + 3001.(-0,5) = -1700,5\,kN/m^2$$





Problem: The weight density (specific weight) of the material used to construct the chimney shown in the figure is $\gamma = 18 kN/m^3$. Find the magnitude of the distributed load q (kN/m^2) such that no tension occurs at the base of chimney.



Çözüm:

$$W = (3,5^2 - 2,5^2).12.18 = 1296\,kN$$

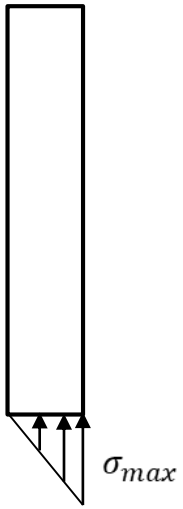
$$M = q.12.3,5.6 = 252\,q\,kNm$$

$$A = 3,5^2 - 2,5^2 = 6\,m^2,\ I_x = \frac{3,5^4 - 2,5^4}{12} = 9,25\,m^4$$

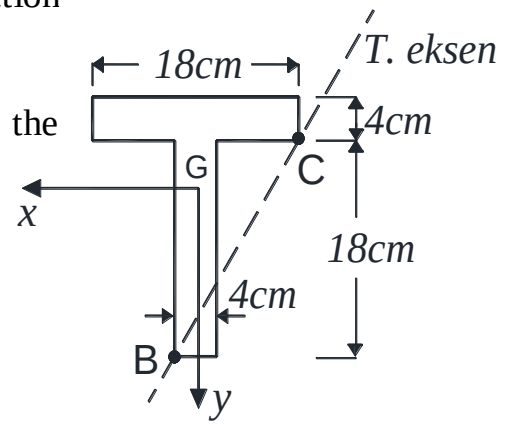
P=18*12*6=1296 kN

For x=1.75m there should be no tension (e.g. $\sigma_{max} = 0$)

$$\sigma_z = \frac{-1296}{6} + \frac{252\,q}{9,25}(1,75) = 0 \ \rightarrow\ q = 4,51 kN/m$$



Problem: Knowing that the neutral axis passes from the corners B and C of the cross-section under the eccentric normal load;
a) Find the application point of the force.
b) If the force is 200 kN tension force, find the max. and min. normal stresses occur at the cross-section and show the stress distribution on the cross-section.



Solution:

$$a)\ y_G = \frac{18.4.9 + 18.4.20}{18.4 + 18.4} = 14,5\,cm$$

$$I_x = \frac{4.18^3}{12} + 4.18.(9-14,5)^2 + \frac{18.4^3}{12} + 18.4.(20-14,5)^2 = 6396\,cm^4$$

$$I_y = \frac{4.18^3}{12} + \frac{18.4^3}{12} = 2040\,cm^4$$

$$B(2,14,5),\quad C(-9;3,5)$$

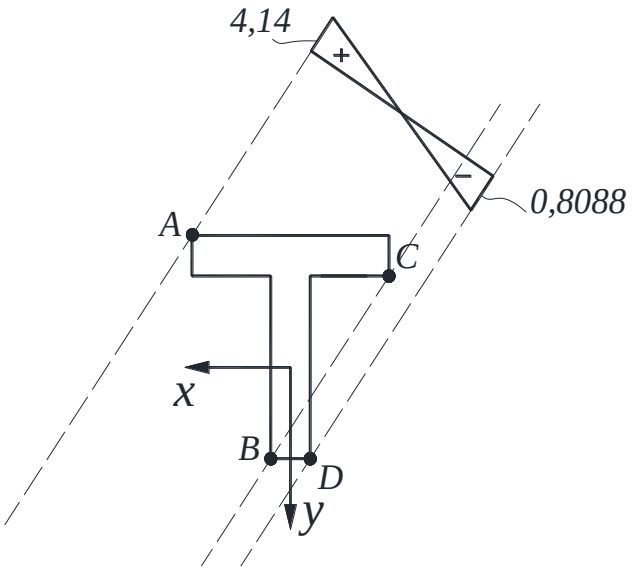
$$A = 2.18.4 = 144\,\text{cm}^2$$

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1) \Rightarrow y = 1,636x + 11,228\ (\text{Neutral axis}),\quad M_x = P.e_y,\ M_y = -P.e_x$$

$$\sigma_z = \frac{P}{144} + \frac{P.e_y}{6396}y + \frac{P.e_x}{2040}x \Rightarrow for\ N.axis \Rightarrow \sigma_z = 0 \Rightarrow \frac{1}{144} + \frac{e_y}{6396}y + \frac{e_x}{2040}x = 0$$

$$y = -\frac{6396}{2040}.\frac{e_x}{e_y}x - \frac{6396}{144e_y} = 1,636x + 11,228 \Rightarrow e_y = -\frac{6396}{144.11,228} \Rightarrow e_y \cong -3,955\,cm$$

$$e_x = \frac{1,636.3,955.2040}{6396} \Rightarrow e_x \cong 2,063\,cm$$



$$b)\ M_x = 200.(-3,955) = -791\,\text{kNcm},\quad A(9;-7,5),\ D(-2;14,5)$$

$$M_y = -200.(2,063) = -412,6\,\text{kNcm}$$

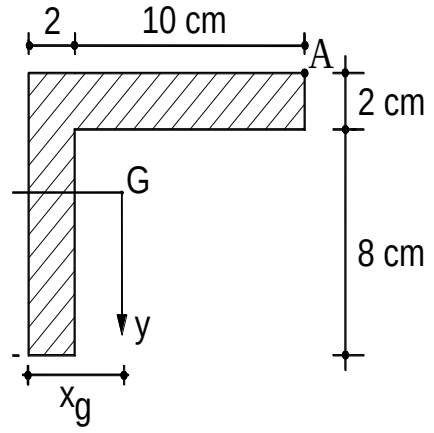
$$\sigma_z = \frac{N}{A} + \frac{M_x}{I_x}y - \frac{M_y}{I_y}x = \frac{200}{144} - \frac{791}{6396}y + \frac{412,6}{2040}x \Rightarrow \sigma_A(9;-7,5) \cong 4,14\ kN/cm^2,\quad \sigma_D(-2;14,5) = -0,8088\ kN/cm^2$$

$$\sigma_z = 0 \Rightarrow y = 1,636x + 11,228\quad (\text{T.E})$$

Problem: P=10000 N compression force is acting at point A of the cross-section in the **Figure 2**.

a) Find the equation of the neutral axis and Show it on the section.

b) Find the max. and min. normal stresses and draw the normal stress diagram over the section.



$x_g = 4\,cm$, $y_g = 7\,cm$

$$\sigma_z = \frac{N}{A} + \frac{M_x I_y + M_y I_{xy}}{I_x I_y - I_{xy}^2} \cdot y - \frac{M_x I_{xy} + M_y I_x}{I_x I_y - I_{xy}^2} \cdot x$$

If the coordinate axes are not principal coordinates

$x_g = 4\,cm$, $y_g = 7\,cm$, $I_x = 333,33\,cm^4$, $I_y = 533,33\,cm^4$, $I_{xy} = 240\,cm^4$, $N = -10000\,N$,

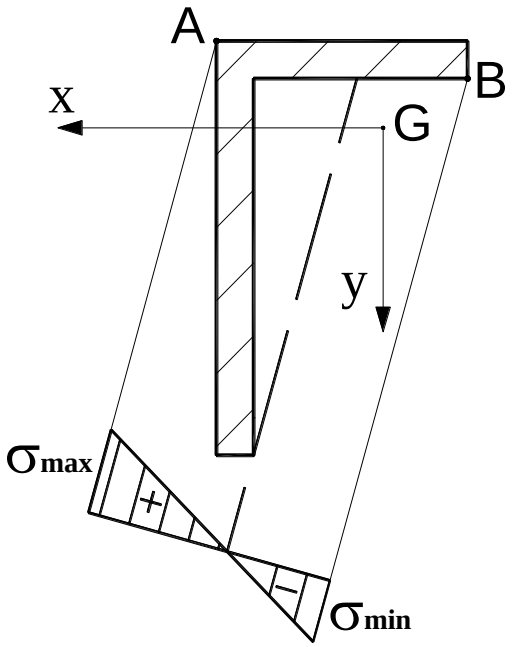
$M_x = 30000\,Ncm$, $M_y = -80000\,Ncm$

$$\sigma_z = -250 - 26,63y + 161,98x = 0 \quad \rightarrow T.E$$

for $x = 0$ for y= 0

$y = -9.39$ $x = 1.5434$

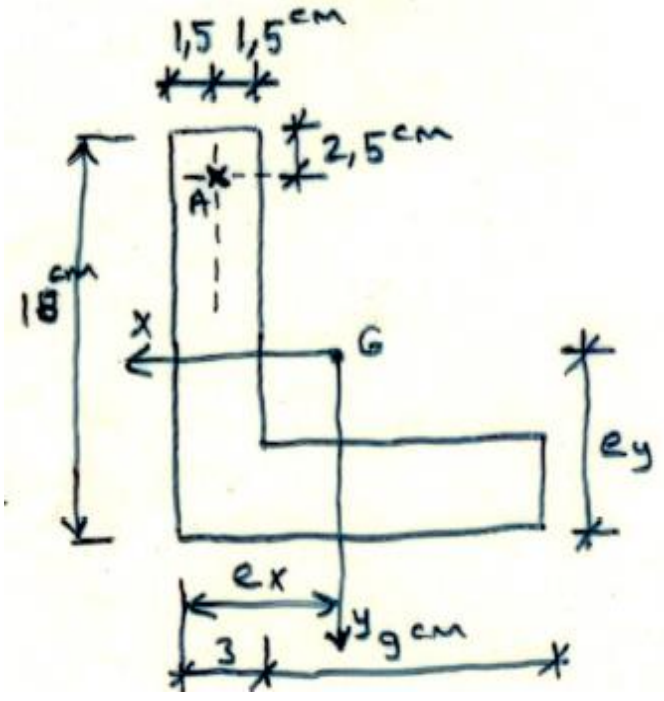
$\sigma_A = 477,81\,N/cm^2 \rightarrow \sigma_{max}$, $\sigma_B = -1519,21\,N/cm^2 \rightarrow \sigma_{min}$



Problem: The cross section of a column with known dimensions is under compression of eccentric load, P as seen in the figure. The allowable tensile stress is $\sigma_{t,all} = 1200 N/cm^2$ and the allowable compressive stress is $\sigma_{c,all} = 6000 N/cm^2$. Find the maximum force P which can be carried by the cross section. Draw the normal stress distribution over the cross section of the column for this obtained force.

$$e_x = \frac{(3.18).1,5 + (9.3).7,5}{3.18 + 9.3} = 3,5 \text{ cm}$$

$$e_y = \frac{(3.18).9 + (9.3).1,5}{3.18 + 9.3} = 6,5 \text{ cm}$$



$$I_x = \frac{1}{12}\cdot 3.18^3 + (3.18).(9-6,5)^2 + \frac{1}{12}\cdot 9.3^3 + (9.3)(6,5-1,5)^2 = 2490,75 \text{ cm}^4$$

$$I_y = \frac{1}{12}\cdot 18.3^3 + (18.3)(3,5-1,5)^2 + \frac{1}{12}3.9^3 + (3.9)(7,5-3,5)^2 = 870,75 \text{ cm}^4$$

$$I_{xy} = (3.18)(6,5-9)(3,5-1,5) + (9.3)(6,5-1,5)(3,5-7,5) = -270 - 540 = -810 \text{ cm}^4 \quad (3)$$

$$A = 3.18 + 9.3 = 81 \text{ cm}^2$$

$A = 3.18 + 9.3 = 81\ cm^2$

$$\sigma_z = \frac{N}{A} + \frac{M_x I_y + M_y I_{xy}}{I_x I_y - I_{xy}^2} \cdot y - \frac{M_x I_{xy} + M_y I_x}{I_x I_y - I_{xy}^2} \cdot x$$

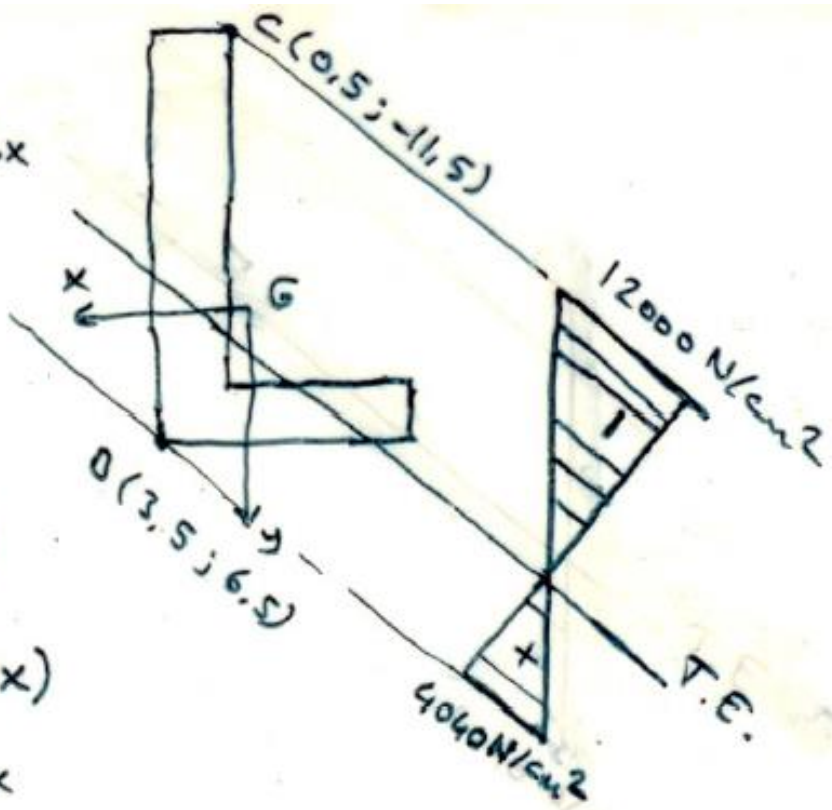
$N = P \ ; \ M_x = 9P \ ; \ M_y = 2P$

$$\sigma_z = P\left(\frac{-1}{81} + \frac{9.870{,}75 + 2.(-810)}{(2490{,}75)(870{,}75) - (-810)} \cdot y - \right.$$

$$\left. - \frac{9(-810) + 2.2490{,}75}{(2490{,}75)(870{,}75) - (-810)^2} \cdot x\right)$$

$\sigma_z = P(-0{,}0123 + 0{,}00411y + 0{,}00153x)$

$T.E. \rightarrow \sigma_z = 0 \rightarrow y = 3 - 0{,}37x$



$\sigma_D = P(-0{,}0123 + 0{,}00411(6{,}5) + 0{,}00153(3{,}5) \leq 6000 \rightarrow P \leq 303490\ N$

$\sigma_C = P(-0{,}0123 + 0{,}00411(-11{,}5) + 0{,}00153(0{,}5)) \leq -12000 \rightarrow P \leq 204080\ N$

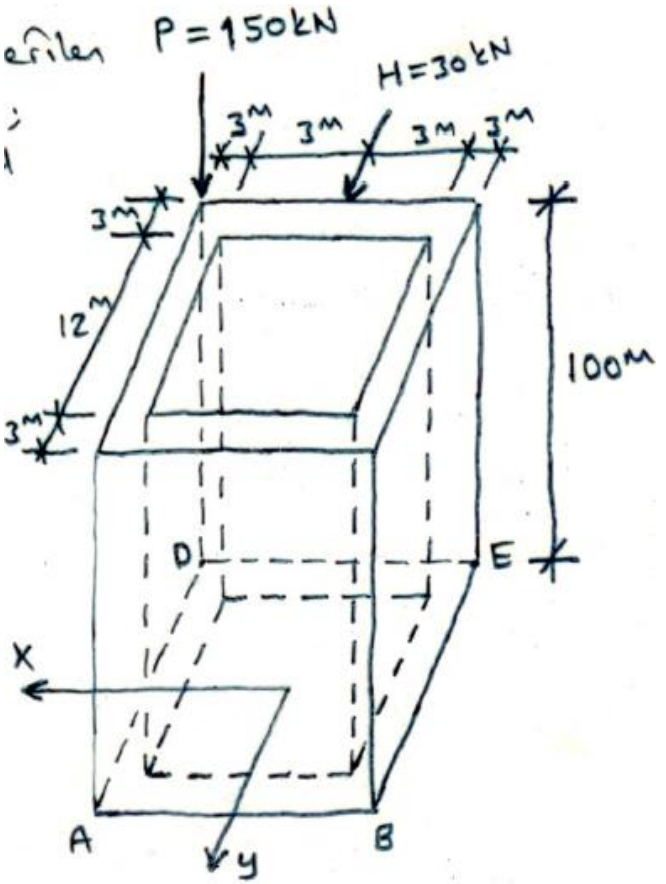
Kısaca olan kuvvet sectik $P \leq 204080\ N$

$\sigma_D = 0{,}01977 . 204080 \rightarrow \sigma_D = 4040\ N/cm^2$

$\sigma_C = -0{,}0588 . 204080 \rightarrow \sigma_C = -12000\ N/cm^2$

Problem: A box column with known dimensions is under the effect of eccentric compression load *P=150 kN* and a load which is perpendicular to the axis of the column *H=30 kN* as seen in the figure.

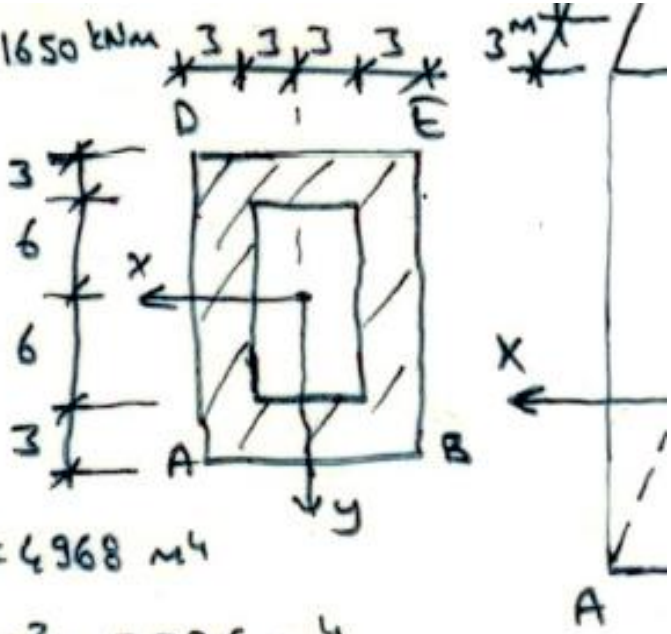
a) Find the normal stresses at the points A, B, D, E,
b) write the equation of the neutral axis and draw the diagram of the normal stress distribution.



$M_x = 150.9 - 30.100 = -1650\ kNm$

$M_y = 150.6 = 900\ kNm$

$A = 12.18 - 6.12 = 144\ cm^2$



$I_x = \frac{1}{12}.12.18^3 - \frac{1}{12}.6.12^3 = 4968\ m^4$

$I_y = \frac{1}{12}.18.12^3 - \frac{1}{12}.12.6^3 = 2376\ m^4$

$$\sigma_z = \frac{N}{A} + \frac{M_x}{I_x} y - \frac{M_y}{I_y} x \rightarrow \sigma_z = \frac{-150}{144} + \frac{-1650}{4968} y - \frac{900}{2376} \cdot x$$

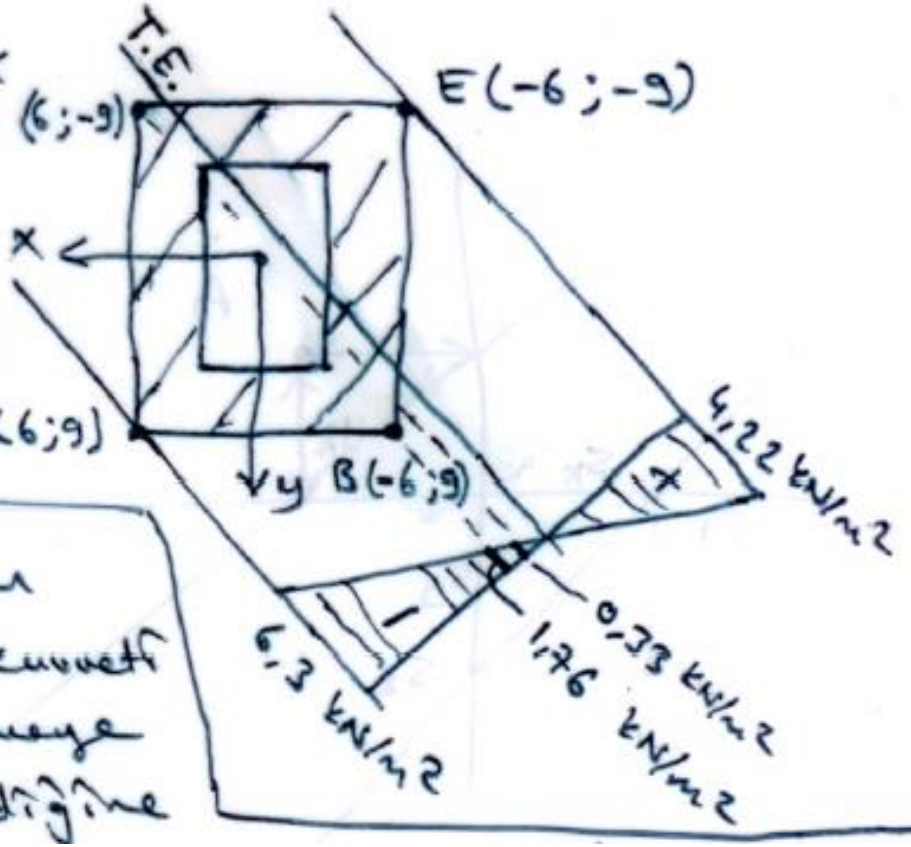
$$\sigma_z = -1,0417 - 0,3321\, y - 0,3788\, x$$

$\sigma_A = -1,0417 - 0,3321 \cdot 9 - 0,3788 \cdot 6 = -6,3\ kN/m^2$

$\sigma_B = -1,0417 - 0,3321 \cdot 9 - 0,3788(-6) = -1,76\ kN/m^2$

$\sigma_D = -1,0417 - 0,3321(-9) - 0,3788 \cdot 6 = -0,33\ kN/m^2$

$\sigma_E = -1,0417 - 0,3321(-9) - 0,3788(-6) = 4,22\ kN/m^2$



Problem 5: Şekildeki gibi üçgen bir kesite C noktasında P kuvveti uygulanmıştır. Malzemenin çekmeye karşı dayanıksız olduğu bilindiğine

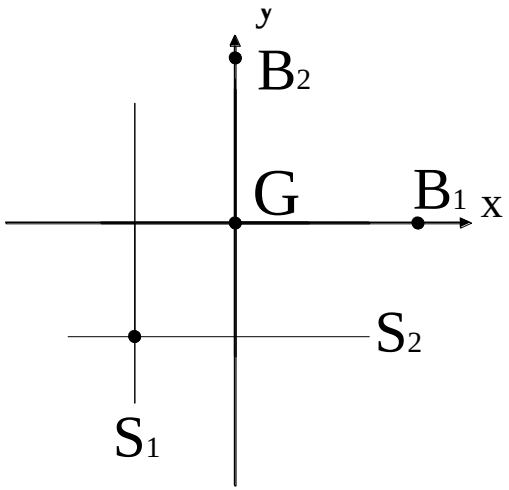
3- Application point of ecc. normal force and Neutral Axis

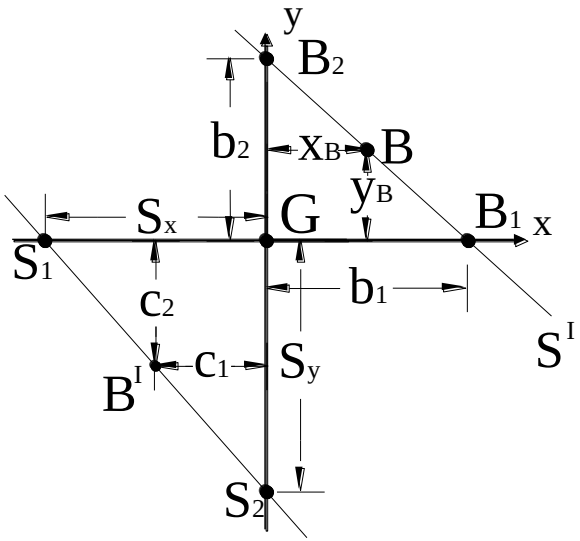
Properties

a) When application point approaches to the center of gravity, neutral axis goes away from it.

b) When application point is on one of the principal axis, which is x_B or y_B equals zero, neutral axis is parallel to the other principal axis

c) If the application points B is on a fixed line, than neutral axes belonging to these application points passes from a fixed point.



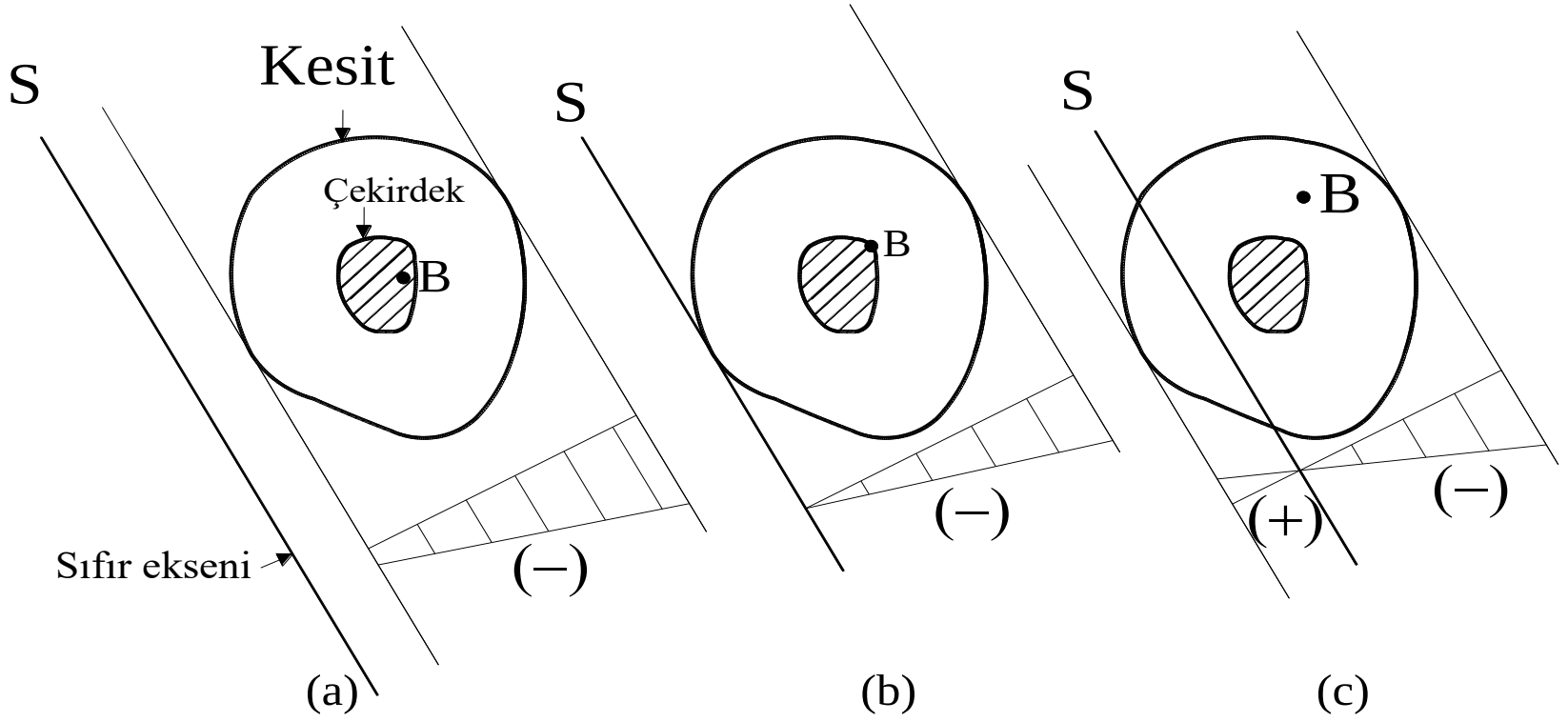


4- Core

For which locations of the eccentric normal force cause all the normal stresses have the same sign ?

Lets determine specific locations for eccentric normal force in which all the neutral axes cut the section and leave the whole section on one side and in the limit case they become tangent to the section. The application points which satisfy these conditions are located in a specific region is called the core of the cross section. For example, let the force acting at the point B be a compressive force. In this case, the position of neutral axis for the forces in the core region, on the border of the core region and outside the core region, are shown below.

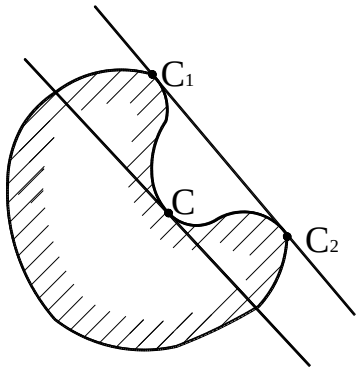
The core is a region inside the cross-section such that the stresses have a negative sign when the application point of the eccentric normal force is inside this region.



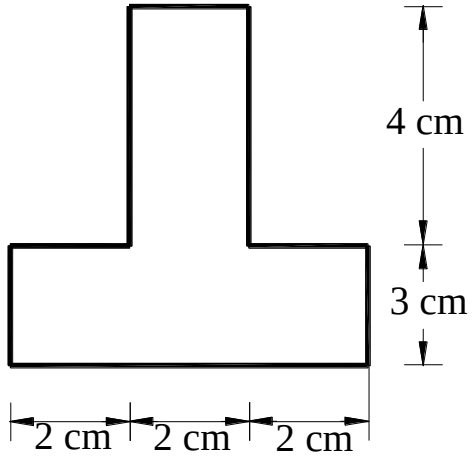
The coordinates of the application points B for which the neutral axes touches the border of the cross section and leaves the cross section at one side can be calculated by formulas $S_x = -i_y^2 / x_B$ and $S_y = -i_x^2 / y_B$ (5).

Moving S as staying tangent to the border of the cross section, the corresponding application points B indicates the perimeter of the core of the cross section.

If the cross-section is not concave, some tangents cut the cross-section area and both sides of the neutral axis have some part of the section. For example, tangent at point C is such a point shown in the figure given below. These types of tangents mustn't be considered in the calculation of the core of the section; because the requirement of having a single sign is not satisfied. In Figure 10, C_1CC2 part of the perimeter will not be considered in the calculation of the core of the section, instead common tangent C1 C2 will be considered. According to this, even if the cross-section is not always concave, the core of the section will always be concave.



Example: **Find the core of the given section.**



$$i_x = \frac{I_x}{A} \; ; \; S_x = -i_y^2 / x_B \; ; \; S_y = -i_x^2 / y_B$$

$$y_g = \frac{6.3.1{,}5 + 2.4.5}{6.3 + 2.4} = \frac{67}{26} = 2{,}58^{cm}$$

$$I_x = \frac{6.3^3}{12} + 6.3\left(1{,}5 - 2{,}58\right)^2 + \frac{2.4^3}{12} + 2.4\left(5 - 2{,}58\right)$$

$$I_x = 20{,}25 + 31{,}5 + 16 + 29{,}04 = 92.02cm^4$$

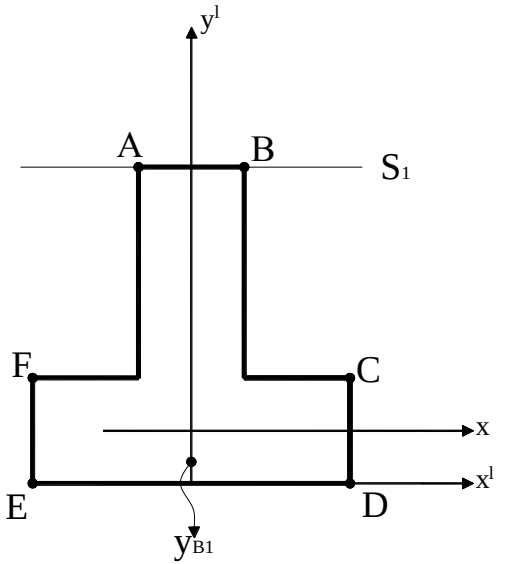
$$I_y = \frac{3.6^3}{12} + \frac{4.2^3}{12} = 56{,}67cm^4$$

$$A = 6.3 + 2.4 = 26cm^2$$

$$i_x^2 = \frac{92{,}02}{26} = 3{,}54cm^2 \quad i_y^2 = \frac{56{,}67}{26} = 2{,}18cm^2$$

$$AB \rightarrow S_{1x} = \infty \; ; \; S_{1y} = 4{,}42$$

$$x_{B1} = \frac{-2{,}18}{\infty} = 0 \; ; \; y_{B1} = -\frac{3{,}54}{4{,}42} = -0{,}8cm$$



$BC \rightarrow S_2$ **equation** $B(+1;+4,42)\ \ C(+3;+0,42)$

B and C are the points on S_2

Equation of S_2 **neutral axis ;** $y_{S_2} = mx_{S_2} + n$

$$B \rightarrow +4,42 = +m.1 + n \quad \textbf{(1)}$$

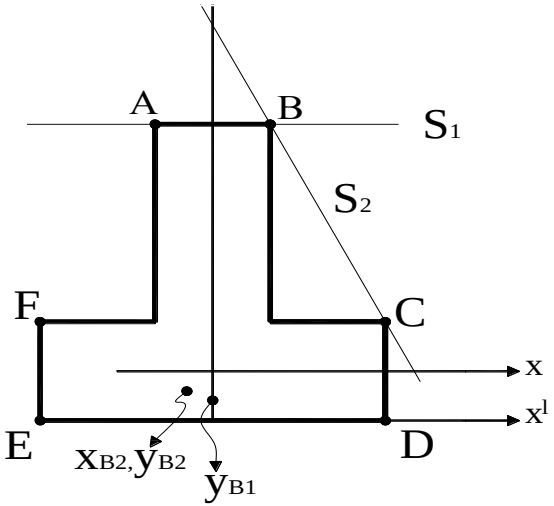
$$C \rightarrow +0,42 = +m.3 + n \quad \textbf{(2)}$$

From (1) and (2) it is found that m=-2, n=6,42.

$$y_{S_2} = -2x_{S_2} + 6,42 \rightarrow S_{2y} = 6,42; y_{s2} = 0 \rightarrow S_{2x} = +3,21$$

$$x_{B2} = -\frac{2,18}{3,21} = -0,679; y_{B2} = -\frac{3,54}{6,42} = -0,551$$

$$\sigma_z(x,y) = 0 \rightarrow \left(1 + \frac{y_B y_{S_2}}{i_x^2} + \frac{x_B x_{S_2}}{i_y^2}\right) = 6,42 - y_{S_2} - 2x_{S_2}$$

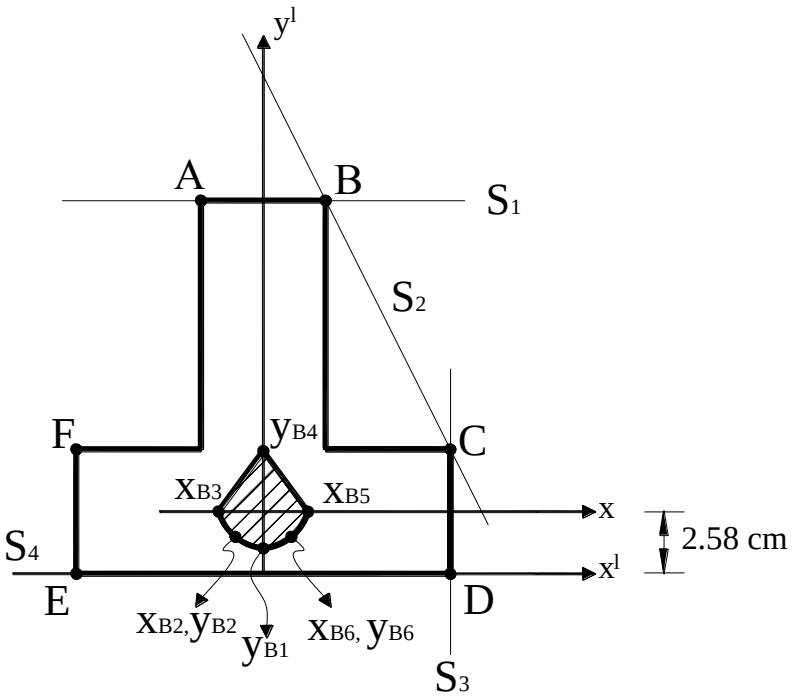


$$CD \rightarrow S_{3x} = 3, S_{3y} = \infty$$

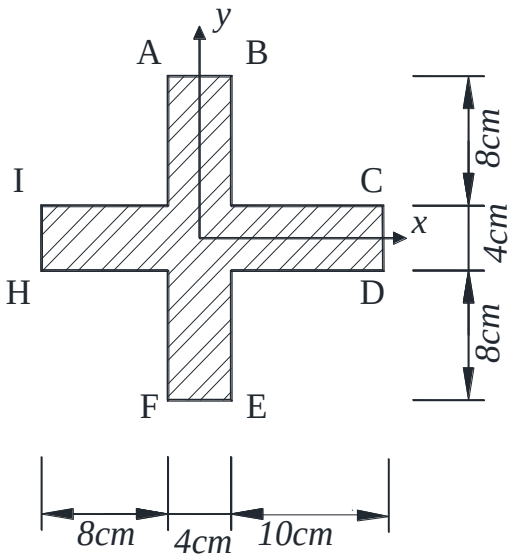
$$x_{B3} = -\frac{2,18}{3} = -0,73; y_{B3} = -\frac{3,54}{\infty} = 0$$

$$DE \rightarrow S_{4x} = \infty, S_{4y} = -2,58$$

$$x_{B4} = -\frac{2,18}{\infty} = 0; y_{B4} = \frac{-3,54}{-2,58} = 1,37$$



Problem: Draw the core of the cross-section given below.



$$x_G = \frac{2.4.8.10 + 22.4.11}{2.4.8 + 22.4} = 10,57\ c$$

$$I_x = \frac{4.20^3}{12} + \frac{8.4^3}{12} + \frac{10.4^3}{12} = 2762,6\ cm^4$$

$$I_y = \frac{4^3.20}{12} + 20.4.\left(10-10,57\right)^2 + \frac{4.10^3}{12} + 4.10.\left(17-10,57\right)^2 + \frac{4.8^3}{12} + 4.8.\left(4-10,57\right)^2 = 3671\ cm^4,$$

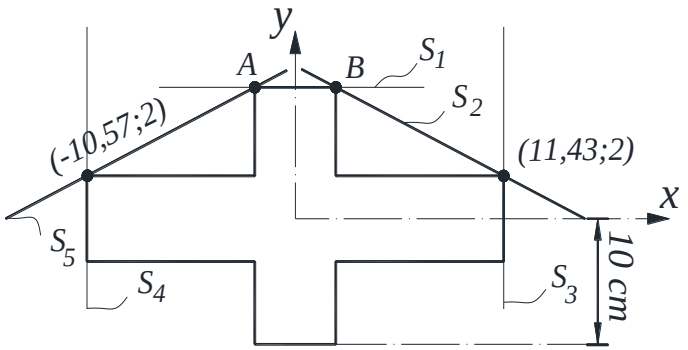
$$A = 152\ cm^2$$

$$i_x^2 = 2762/152 = 18,17\ cm, \quad i_y^2 = 3671/152 = 24,15\ cm^4$$

If S_1 **is neutral axis,**

$$S_{1x} = \infty, \quad S_{1y} = 10$$

$$x_{B1} = 0, \quad y_{B1} = -18,17/10 = -1,817$$



If S_2 is neutral axis,

$$y = -0,8x + 11,144$$

$$S_{2x} = 13,93, \quad S_{2y} = 11,144 \Rightarrow x_{B2} = -24,15/13,93 = -1,73 \quad y_{B2} = -18,17/11,144 = -1,63$$

If S_3 is neutral axis,

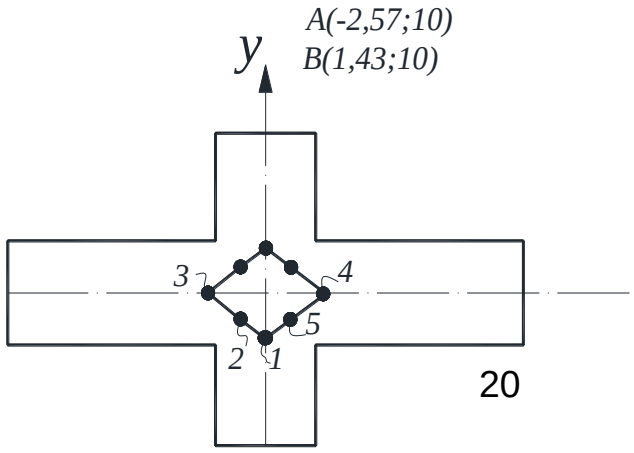
$$S_{3x} = 11,43; \quad S_{3y} = \infty$$

$$x_{B3} = -2,11 \quad y_{B3} = 0$$

If S_4 is neutral axis,

$$S_{4x} = 10,57; \quad S_{4y} = \infty$$

$$x_{B4} = 2,284 \quad y_{B4} = 0$$



If S_5 is neutral axis,

$$y = x + 12,57$$

$$S_{5x} = -12,57, \quad S_{5y} = 12,57 \Rightarrow x_{B5} = 1,92\ cm, \quad y_{B5} = -1,445\ cm$$

MATERIAL WHICH IS WEAK IN TENSILE (or Tensionless)

If the applied force is outside of the core, the stress of the cross-section is double-signed. If the material is weak in tensile, the tensile stresses shouldn't be considered while calculating the stresses:

- **The compression region and the region without stress are divided by a straight line which is called neutral axis or zero axis.**
- **Compression stresses of the cross-section increase linearly with respect to the increasing distance from the neutral axis.**

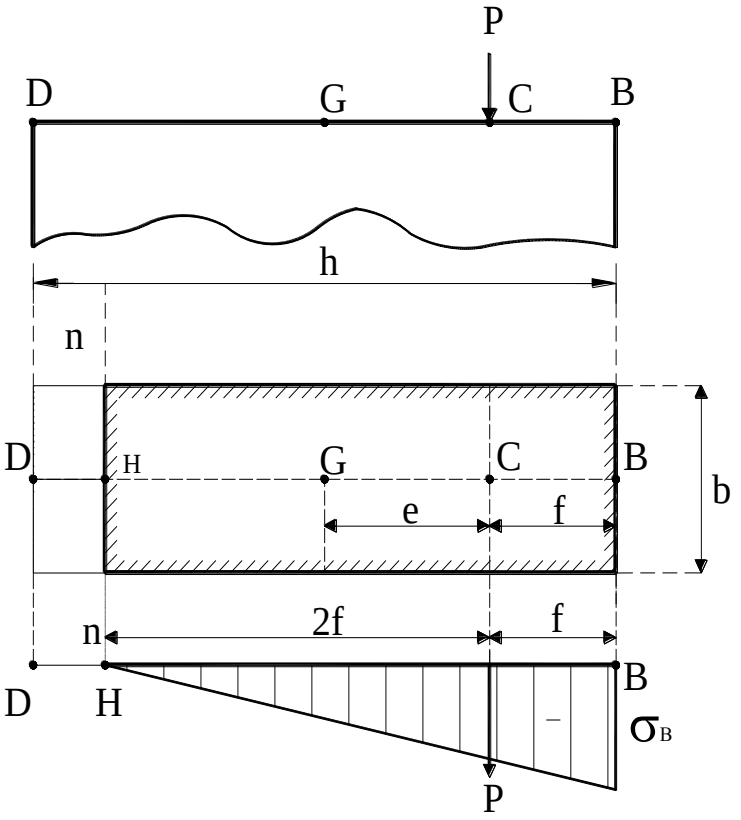
If the application point is outside of the core, by using the assumptions given above and equilibrium equations $\sum F_y = 0$, $\sum M = 0$, **the location of the neutral axis and compression stresses can be defined.**

For the material which is weak in tensile and has a rectangular cross section, (figure 11); when the force is on the x axis which is the symmetry axis for the cross-section, then the eccentricity is also on the x axis and it is explicit that the neutral axis is parralel to the y axis.

If the cross-section has't got a symmetry with respect to the axis which passes from the application point of the force, the calculation proses must be trial and error and because of this the solution will be too long. The resultant of stress prism must be at the point C because of equilibrium necessities for the rectangular section. From this condition we get

$$\frac{3f\,b\,\sigma_B}{2} = P \rightarrow \sigma_B = -2\frac{P}{3bf} \qquad (15)$$

Concrete can not withstand tensile forces. Also the foundations of the structures are put on the ground freely and therefore they only can transmit compressive forces and can't transmit the tensile forces to the ground.

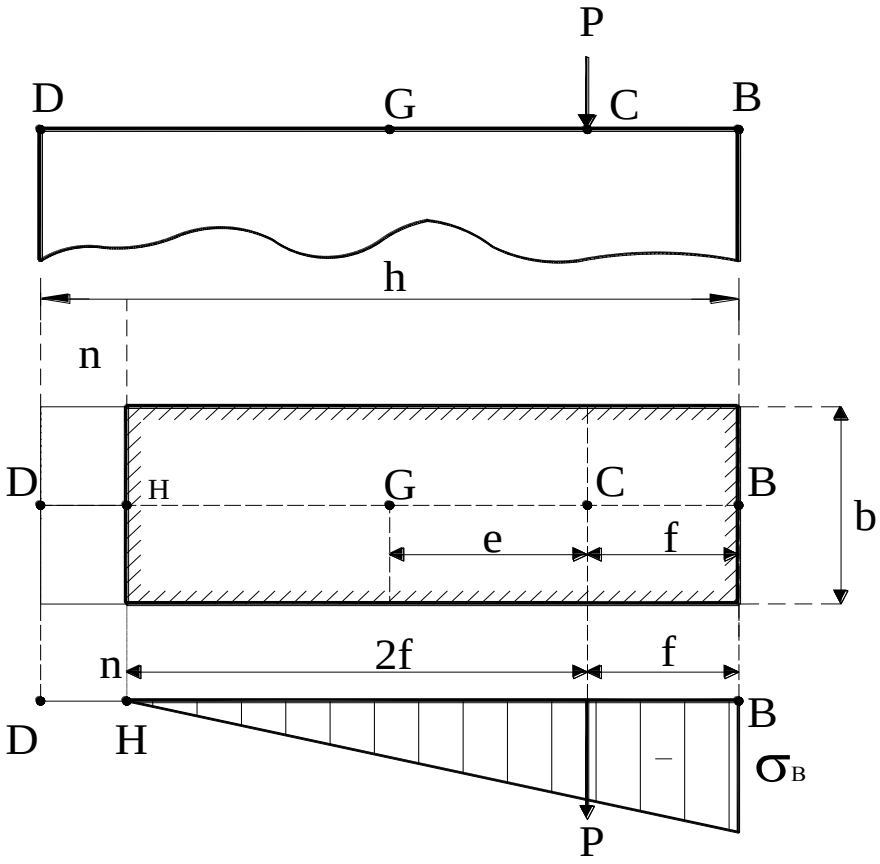


It can be seen from figure that $f = \frac{h}{2} - e$

If $h/2 - e$ is substituted for f, then the stress equation can be be written as

$$\sigma_B = -2\frac{4P}{3bh(1-2e/h)} \qquad (16)$$

It can be seen that this equation is valid for $(h/6 \le e < h/2)$. If If e<h/6, then the stress distribution becomes trapezoidal, therefore Equation (16) is not valid any more, and when e>h/2 then the stress becomes infinite. If we omit the tensile stresses For the material capable of carrying tensile stresses, for the investigated example, the stres distribution is double-signed and the maximum compression stress σ'_B is



$$\sigma' = \frac{P}{A} + \frac{M}{I}y_{\max} = \frac{P}{bh} + \frac{Pe}{bh^3/12}\frac{h}{2} \rightarrow \sigma'_B = \frac{P(e+h/6)}{3bh^2/6} \qquad (17)$$

The ratio between the resulting stresses from the two theories is written as

$$\frac{\sigma_B}{\sigma'_B} = \frac{4}{(3-6e/h)(1+6e/h)} \ge 1 \quad (h/6 \le e < h/2) \qquad (18).$$

for $e = h/2 \;\; \sigma_B \rightarrow \infty$.

It is possible to generalize this theory for any cross-section. Consider a cross-section that has symmetry with respect to the plane of forces. Let's consider that the compression force P is applied at the point C which is outside of the core, and the corresponding neutral axis is n-n. In the figure compression effective region is hatched: the other region is not effective because of the tension weakness of the material that means there is no stress in this region.

For the linear law, the stress at the distance (u) from the neutral axis is $\sigma = k.u$ **(19)** .

A` = effective area of the section

S_n = static moment with respect to the neutral axis n-n

I_n = moment of inertia with respect to theneutral axis n-n

k = proportion coefficient

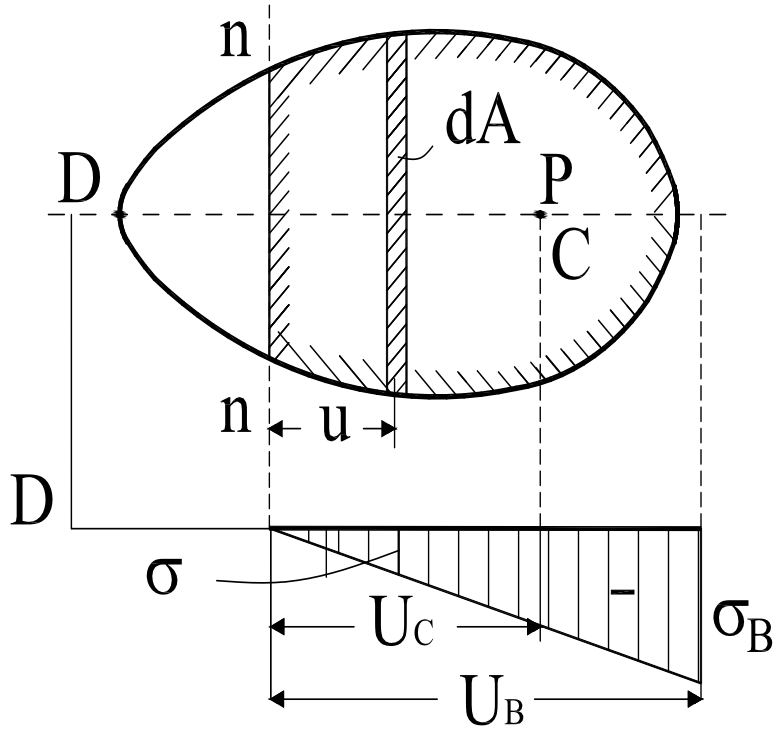
$$-P = \iint_{A'} \sigma.dA \quad -P.U_C = \iint_{A'} u.\sigma.dA \qquad (20)$$

$$-P = k\int_0^{U_B} u.dA = k.S_n \rightarrow k = \frac{-P}{S_n} \qquad (21a)$$

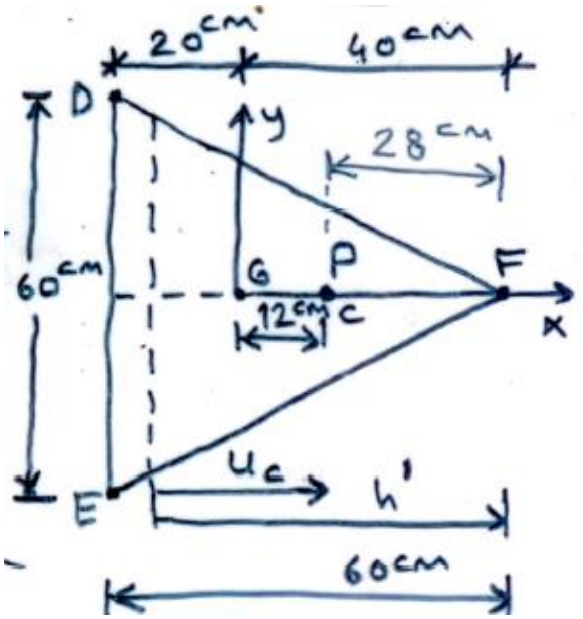
$$-P.U_C = k\int_0^{U_B} u^2.dA. = k.I_n \qquad (21b)$$

$$-P.U_C = k\ I_n = -\frac{P}{S_n} I_n \rightarrow U_C = \frac{I_n}{S_n} \qquad (22)$$

$$\sigma = k.u = \frac{-P.u}{S_n} \quad ve \quad \sigma_B = \frac{-P.U_B}{S_n} \qquad \textbf{(23)}$$



Problem: The force P is applied perpendicularly on an isosceles triangle at the point C. It is known that the materail is tensionless. Namely it can not withstand tensile stresses. Find the location of the neutral axis.



Solution: First of all we will check whether or not the application point C of the force is in the core of the section.

$$s_x = -\frac{i_y^2}{x_c}\ ;\ i_y^2 = \frac{I_y}{A}\ ;\ A = \frac{bh}{2} = \frac{60.60}{2} = 1800\,cm^2$$

$$I_y = \frac{bh^3}{36} = \frac{60.60^3}{36} = 360000\,cm^4$$

$$i_y^2 = \frac{360.000}{1800} = 200\,cm^2$$

By assuming that S_C is tangent to DE, we obtain $S_{Cx} = -20cm$. In this case $x_C = -i_y^2 / S_{Cx} = -200/(-20) = 10cm$. $10cm < |GC| = 12cm$. Therefore the stress distribution has double sign. It means that the application point is out of the core. We can find the location of the neutral axis by trial and error. Lets assume that the neutral axis is at the left side of the point G and the distance between the neutral axis and the point F is h' and the point C is U_C.

1. Adım $h' = 57 \rightarrow I_n = \frac{1}{12} \cdot 57.57^3 = 879666{,}75\,cm^4$; $S_n = \frac{57.57}{2} \cdot \frac{57}{3} = 30865{,}5\,cm^3$

$$u_c = \frac{I_n}{S_n} = \frac{879666{,}75}{30865{,}5} = 28{,}5\,cm \neq u_c = 57 - 28 = 29\,cm$$

2. Adım $h' = 55 \rightarrow \frac{I_n}{S_n} = \frac{55.55^3}{12} \cdot \frac{6}{(55.55.55)} = 27{,}5 \neq u_c = 55 - 28 = 27\,cm$

1. Adım $h' = 57 \rightarrow I_n = \frac{1}{12} \cdot 57.57^3 = 879666{,}75\ cm^4$; $S_n = \frac{57.57}{2} \cdot \frac{57}{3} = 30865{,}5\ cm^3$

$$U_c = \frac{I_n}{S_n} = \frac{879666{,}75}{30865{,}5} = 28{,}5\ cm \neq U_c = 57 - 28 = 29\ cm$$

2. Adım $h' = 55 \rightarrow \frac{I_n}{S_n} = \frac{55.55^3}{12} \cdot \frac{6}{(55.55.55)} = 27{,}5 \neq U_c = 55 - 28 = 27\ cm$

3. Adım $h' = 56$ $\frac{I_n}{S_n} = \frac{56.56^3}{12} \cdot \frac{6}{56.56.56} = 28 = U_c = 56 - 28 = 28\ cm$

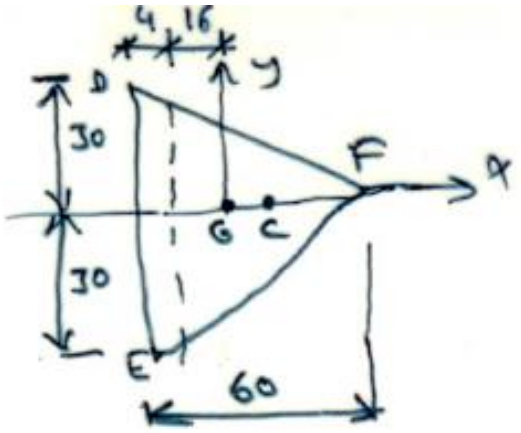
Neutral axis is 56-40=16cm to the left side of the point G for the force P.

Problem: If we invert the problem, namely if the location of the neutral axis is known, then what is the application point of the force $P = 5000\,N$ and the maximum value of the compresive stress?

Solution:

$$-P.U_C = k\ I_n = -\frac{P}{S_n} I_n \rightarrow U_C = \frac{I_n}{S_n} \quad (22)$$

$$\sigma = k.u = \frac{-P.u}{S_n} \quad ve \quad \sigma_B = \frac{-P.U_B}{S_n} \quad (23)$$



$$u_c = \frac{I_n}{S_n} = \frac{56.56^3}{12} \cdot \frac{6}{56.56.56} = 28$$

$$\sigma_B = -\frac{5000.6.56}{56.56.56} = -9{,}57\ kN/m^2$$

em S

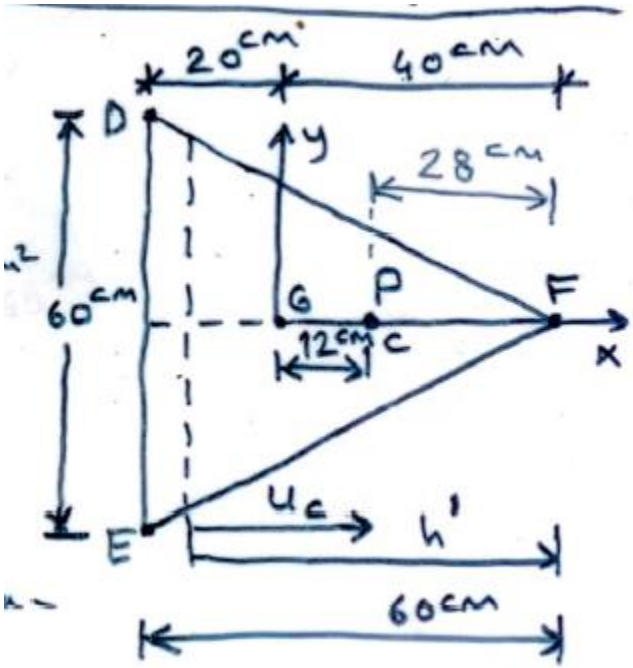
$$u_c = \frac{I_n}{S_n} = \frac{\frac{h' \cdot h'^3}{12}}{\frac{h' \cdot h'}{2} \cdot \frac{h'}{3}} = \underbrace{h' - 2\delta}_{u_c}$$

$$h' - 2\delta = \frac{h'^4 \cdot 6}{12 \cdot h'^3} = \frac{h'}{2}$$

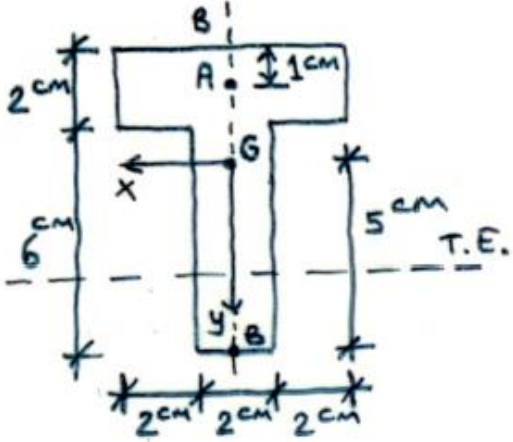
$$h' - 2\delta = \frac{h'}{2} \rightarrow (h' - 2\delta) \cdot 2 = h'$$

$$2h' - 2 \cdot 2\delta = h'$$

$$h' = 2 \cdot 2\delta \rightarrow h' = 56 \text{ cm}$$



Problem: The force P is applied perpendicularly on the given cross section at the point A. It is known that the materail is tensionless. Namely it can not withstand tensile stresses. Also we know that the allowable compressive stress is $\sigma_{c,\,all} = 5000\,N/cm^2$. a) Find the location of the neutral axis. b) Find the maximum force P which can be carried by the cross section.



$$y_G = \frac{2.6.3 + 6.2.7}{2.6 + 6.2} = 5\ cm \qquad A = 2.6 + 6.2 = 24\ cm^2$$

$$I_x = \frac{2.6^3}{12} + 2.6(5-3)^2 + \frac{6.2^3}{12} + 6.2.(7-5)^2 = 36+48+4+48 = 136\ cm^4$$

Tarafsız eksen : $S_y = -\frac{i_x^2}{y_A}$; $i_x^2 = \frac{I_x}{A}$; $S_y = -\frac{136}{24}\cdot\frac{1}{-2} = 2,83\ cm$

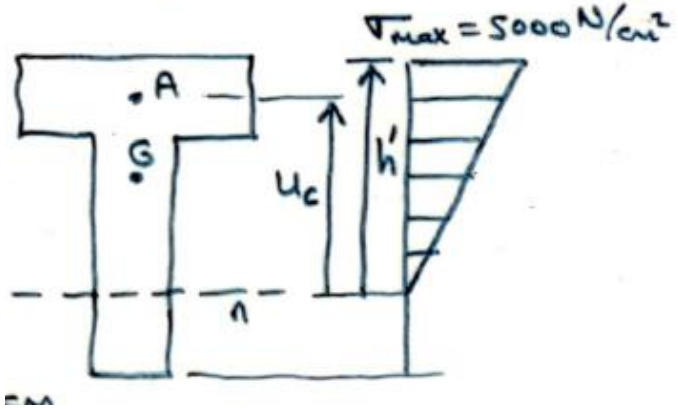
According to this result, the neutral axis is between the points G and B. Hence the stress distribution has double sign. It means that the application point is out of the core. Also, we know that the material is tensionless. Namely there can't be double sign in the cross section. In this case, we have to determine the new stress distribution which has only one sign.

1. Adım $h' = 6$; $I_n = \frac{2.4^3}{3} + \frac{6.2^3}{12} + 6.2.5^2$

$$I_n = 42,67 + 4 + 300 = 346,67\ cm^4$$

$$S_n = 6.2.5 + 2.4.2 = 60 + 16 = 76\ cm^3$$

formülden $u_c = \frac{I_n}{S_n} = \frac{346,67}{76} = 4,56\ cm$; Geometriden $u_c = 6 - 1 = 5\ cm$



2. Adm $h' = 4$ $\quad I_n = \frac{2.2^3}{3} + \frac{6.2^3}{12} + 6.2.3^2 = 5,33 + 4 + 108 = 117,33 \text{ cm}^4$

$S_n = 6.2.3 + 2.2.1 = 36 + 4 = 40 \text{ cm}^3$

$\frac{I_n}{S_n} = \frac{117,33}{40} = 2,93 \qquad u_c = 4 - 1 = 3 \text{ cm}$

3. Adm $h' = 3,5$ $\quad I_n = \frac{2.(1,5)^3}{3} + \frac{6.2^3}{12} + 6.2.2,5^2 = 2,25 + 4 + 75 = 81,25 \text{ cm}^4$

$S_n = 6.2.2,5 + 2.1,5.0,75 = 30 + 2,25 = 32,25 \text{ cm}^3$

$\frac{I_n}{S_n} = \frac{81,25}{32,25} = 2,52 \qquad u_c = 3,5 - 1 = 2,5 \text{ cm}$

4. Adm $h' = 3,61$ $\quad I_n = \frac{2.(1,61)^3}{3} + \frac{6.2^3}{12} + 6.2.2,61^2 = 2,78 + 4 + 81,75 = 88,53$

$S_n = 6.2.2,61 + 2.1,61.\frac{1,61}{2} = 31,32 + 2,59 = 33,91 \text{ cm}^3$

$\frac{I_n}{S_n} = \frac{88,53}{33,91} = 2,61 \qquad u_c = 3,61 - 1 = 2,61$

Tamam

$\sigma_B = -\frac{P.u_B}{S_n}$

$\sigma_B = -5000 = -\frac{P.3,61}{33,91} \longrightarrow P = 46966,78 \text{ N}$

Second solution way: This problem could be solved directly by means of the stresss distribution and the equilibrium equations.

Problem 6

$$u_c = \frac{I_n}{S_n} =$$

$$I_n = \frac{6.2^3}{12} + 6.2.(h'-1)^2 + \frac{2.(h'-2)^3}{3}$$

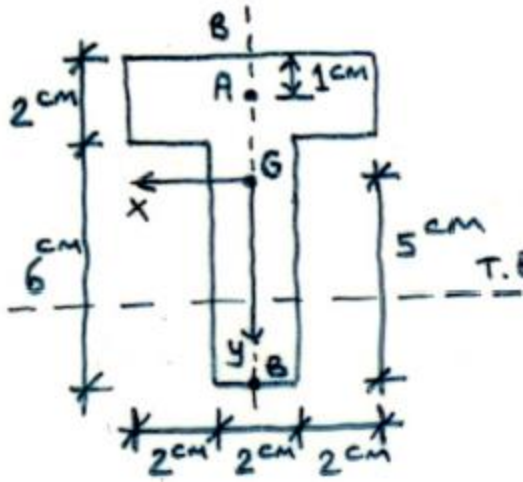
$$I_n = \frac{48}{12} + 12(h'-1)^2 + \frac{2.(h'-2)^3}{3}$$

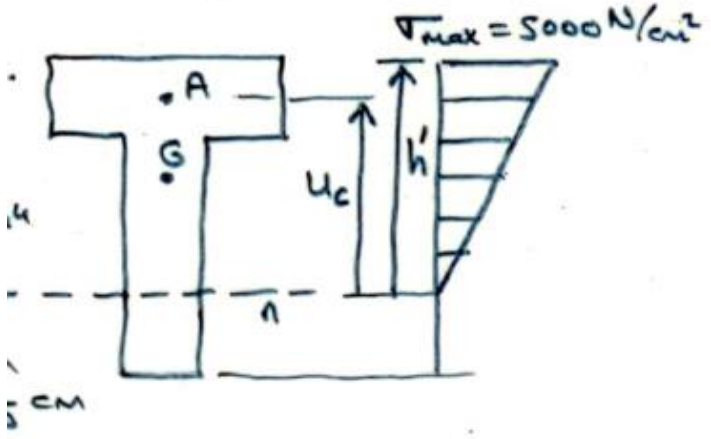
$$S_n = 6.2.(h'-1) + \frac{2.(h'-2).(h'-2)}{2}$$

$$S_n = 12(h'-1) + (h'-2)^2$$

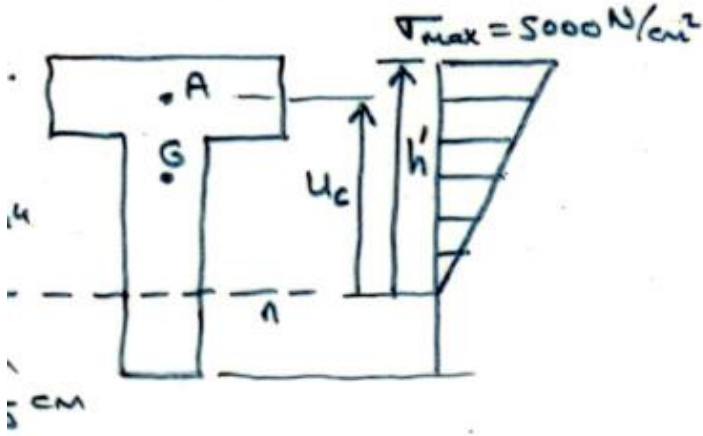
$$S_n = 12h' - 12 + h'^2 - 4h' + 4$$

$$\boxed{S_n = h'^2 + 8h' - 8}$$





$$u_c = \frac{4 + 12(h'-1)^2 + \frac{2}{3}(h'-2)^3}{12(h'-1) + (h'-2)^2} = h'-1$$



$$4 + 12(h'-1)^2 + \frac{2}{3}(h'-2)^3 =$$

$$12(h'-1)^2 + (h'-1)(h'-2)^2$$

$$4 + \frac{2}{3}(h'-2)^3 - (h'-1)(h'-2)^2 = 0$$

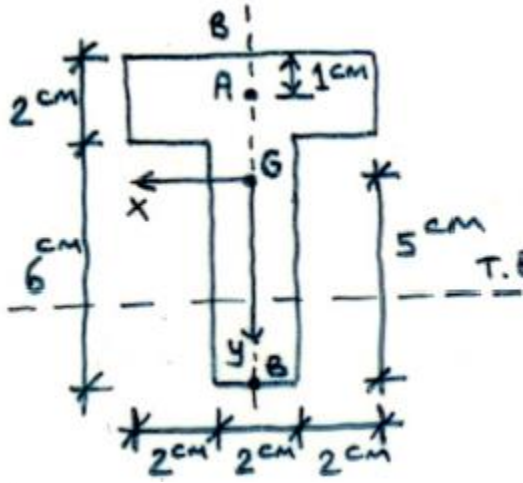
$h' = 3{,}61$ bulunur.

$$4 + \frac{2}{3}(h'-2)^3 - (h'-1)(h'^2 - 4h' + 4) = 0$$

$$4 + \frac{2}{3}(h'-2)^3 - (h'^3 - 4h'^2 + 4h' - h'^2 + 4h' - 4) = 0$$

$$4 + \frac{2}{3}(h'-2)^3 - (h'^3 - 5h'^2 + 8h' - 4) = 0$$

$$4 + \frac{2}{3}(h^3 - 6h^2 + 12h - 8) - (h^3 - 5h^2 + 8h - 4) = 0$$



$$h + \frac{2}{3}h^3 - \frac{12}{3}h^2 + \frac{24}{3}h - \frac{16}{3} - h^3 + 5h^2 - 8h + 4h = 0$$

$$-\frac{16}{3} + 8 + \left(\frac{2}{3} - \frac{3}{3}\right)h^3 + \left(-\frac{12}{3} + \frac{15}{3}\right)h^2 + \left(\frac{24}{3}h - \frac{24}{3}\right)h'$$

$$\frac{24}{3} - \frac{16}{3} = \frac{8}{3}$$

$$-\frac{1}{3}h^3 + \frac{3}{3}h^2 + \frac{8}{3} = 0$$

$$= \frac{1}{3}h + \cdots = 0$$

$$-\frac{1}{3}h^3 + 1h^2 + \frac{0}{3} = 0$$

$$-h^3 + 3h^2 + 0 = 0$$

$$h = 3{,}61289$$